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Microporous MOF for Simultaneous High Thermodynamic and Kinetic Synergistic Separation of Propylene and Propane

Xue Wang^{1#}, Lingxiang Bao^{2,3#}, Jing-Hong Li⁴, Tao Zhao¹, Libo Li⁵, Shuki Torii⁶, Hongwei Chen^{5,7}, Takashi Kamiyama^{2,3}, Ping Miao^{2,3*}, Rui-Biao Lin^{4*}, Banglin Chen^{8,9*}, Junkuo Gao^{1*}

¹China-Uzbekistan Joint Laboratory on Advanced Porous Materials, State Key Laboratory of Bio-based Fiber Materials, School of Materials Science and Engineering, Zhejiang Sci-Tech University, Hangzhou 310018, China

²Spallation Neutron Source Science Center, Dongguan 523803, China

³Institute of High Energy Physics, Chinese Academy of Sciences, Beijing 100049, China

⁴MOE Key Laboratory of Bioinorganic and Synthetic Chemistry, GBRCE for Functional Molecular Engineering, School of Chemistry, IGCME, Sun Yat-Sen University, Guangzhou 510275, China

⁵College of Chemistry and Chemical Engineering, Taiyuan University of Technology, Taiyuan 030024, China

⁶Institute of Materials Structure Science, High Energy Accelerator Research Organization (KEK), Tokai, Ibaraki 319-1106, Japan

⁷State Key Laboratory of Porous Materials for Separation and Conversion, Southwest Institute of Chemical Co., Ltd., Chengdu 610225, China

⁸Fujian Provincial Key Laboratory of Polymer Materials, College of Chemistry and Materials Science, Fujian Normal University, Fuzhou 350007, China

⁹Key Laboratory of the Ministry of Education for Advanced Catalysis Materials, College of

Chemistry and Materials Science, Zhejiang Normal University, Jinhua 321004, China.

*Corresponding author:

Ping Miao, Email: miaoping@ihep.ac.cn; Rui-Biao Lin, Email: linruibiao@mail.sysu.edu.cn;

Banglin Chen, Email: banglin.chen@fjnu.edu.cn; Junkuo Gao, E-mail: jkgao@zstu.edu.cn;

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The authors contribute equally.

Abstract: Efficient separation of propylene from propane is a critical yet challenging industrial process. While rigid molecular sieves offer ideal selectivity, their narrow nanopores inherently constrain adsorption capacity and diffusion kinetics due to compromised thermodynamic-kinetic trade-offs. To address this, we report ZSTU-10, a molecular sieve constructed via structure-directing agents. Uniquely, ZSTU-10 features localized sieving gates for selective guest admission, expansive diffusion channels for rapid transport, and central pore cavities for high-capacity storage. This gate-channel-cavity architecture enables the precise exclusion of propane while facilitating the dense packing and fast diffusion of propylene, achieving simultaneous thermodynamic-kinetics optimization in molecular sieving. Static adsorption experiments demonstrate an exceptional propylene uptake ($97.7 \text{ cm}^3 \text{ cm}^{-3}$) at 298 K and 1 bar. Time-dependent uptake kinetics revealed a propylene diffusion coefficient ($4.29 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$) in ZSTU-10 surpassing benchmarks by two orders of magnitude. Dynamic breakthrough experiments demonstrate that ZSTU-10 produces high-purity propylene (99.1%) with a productivity of 37.5 L kg^{-1} in a single adsorption-desorption cycle.

Introduction

Propylene is one of the most crucial olefins in the petrochemistry industry that serves as a primary feedstock for manufacturing various consumer products and chemicals^{1,2}. Industrial propylene (C_3H_6) is mainly produced via steam cracking of naphtha or fluid catalytic cracking by hydrocarbons, with propane (C_3H_8) as a by-product³⁻⁵. The established industrial route to obtain high-purity C_3H_6 (> 99.5%) from $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixtures relies on traditional thermal-driven cryogenic distillation, which is highly energy intensive separation due to the small volatility difference between the two molecules⁶⁻⁹. Therefore, it is imperative to develop alternative separation technologies with lower energy input and reduced capital cost. Porous materials including zeolites¹⁰⁻¹⁶, activated carbons^{17,18}, and metal-organic frameworks (MOFs)¹⁹⁻²¹ have garnered considerable attention as non-thermal separation platforms, offering the potential to markedly reduce energy consumption relative to cryogenic distillation. However, conventional

zeolites and activated carbons often face limitations in simultaneously achieving high selectivity, high working capacity, and fast mass transport for the challenging C_3H_6/C_3H_8 mixtures, owing to restricted pore-size tunability or broad pore-size distributions. In contrast, MOFs provide highly programmable pore environments and precisely engineerable adsorption sites, enabling rational design of molecular-sieving apertures and host-guest interactions, which has led to their extensively studied for olefin/alkane separation²²⁻²⁶.

MOFs utilizing thermodynamic separation principles exploit differences in adsorption affinity (e.g., via open metal sites) to achieve high equilibrium selectivity^{27,28}. However, the strong host-guest interactions in such materials often necessitate substantial energy for regeneration. In contrast, kinetic separations rely on the differences in diffusion rates and typically feature tuned pore apertures and smooth surfaces to minimize transport resistance²⁹⁻³². Consequently, this approach often suffers from weak host-guest confinement, failing to discriminate between closely related alkanes/alkenes and resulting in low selectivity. While molecular sieving MOFs achieve ultra-high selectivity as an extreme form of kinetic separation, enabled by precise size discrimination within small-sized guests³³⁻³⁶. Yet, they face an inherent structural paradox that the narrowly confined channels intrinsically limit adsorbate diffusion kinetics and reduce accessible adsorption capacity^{37,38}. Furthermore, while elevated temperatures can enhance molecular mobility to mitigate diffusion constraints, they simultaneously reduce adsorption capacity due to exothermic thermodynamic effects. Resolving incompatibilities between pore size and pore shape of MOFs and the optimal operational conditions to achieve synergistic thermodynamic-kinetic separation remains a fundamental challenge³⁹⁻⁴². To address these constraints, we develop a solvent-templated nanoconfinement strategy that precisely tailors MOF pore apertures (in the 3-5 Å regime) and optimizes process conditions (e.g., temperature) to achieve a thermodynamically-kinetically balanced separation. The resulting framework integrates localized molecular sieving gate to selective guest admission, interconnected auxiliary large channels that function as bridge-mediated guest diffusion pathways, and central cavity acting as high-capacity storage reservoirs. This gate-channel-cavity architecture cooperatively enhances diffusion kinetics, guest storage capacity, and host-guest binding affinities (Fig. 1).

Herein, using N-butyl alcohol/N-propyl alcohol /Isopropyl alcohol (*n*-BuOH/*n*-PrOH/*i*-PrOH) as solvent templates, we constructed three isorecticular ultra-microporous MOFs (denoted ZSTU-10, ZSTU-11, and ZSTU-12). These isostructural MOFs feature central pore cages radially connected to constricted gate apertures via diffusion channels, forming an interpenetrated three-dimensional (3D) channel system. Notably, the effective gate aperture of ZSTU-10 precisely interposes between the minimum cross-sections of C₃H₆ ($3.8 \times 4.1 \text{ \AA}^2$) and C₃H₈ ($3.3 \times 4.2 \text{ \AA}^2$), enabling molecular sieving via size exclusion. Gas adsorption analyses reveal a high volumetric C₃H₆ uptake ($97.7 \text{ cm}^3 \text{ cm}^{-3}$) and a notable combined C₃H₆/C₃H₈ equilibrium-kinetic selectivity of 2×10^3 . The C₃H₆ diffusion coefficient ($4.29 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$) rivals benchmark ZU-609 material, while the pronounced C₃H₆/C₃H₈ diffusion-rate differentiation is attributed to the cooperative effect of the localized sieving gate and the short, low-resistance diffusion pathways. Dynamic breakthrough experiments reveal that ZSTU-10 is capable of obtaining high-purity C₃H₆ (99.1%) with a productivity of 37.5 L kg^{-1} . Crystallography studies and neutron powder diffraction experiments confirm this unique framework of ZSTU-10 overcomes intrinsic constraints to achieve high adsorption capacity, rapid diffusion kinetics, and ideal host-guest interactions.

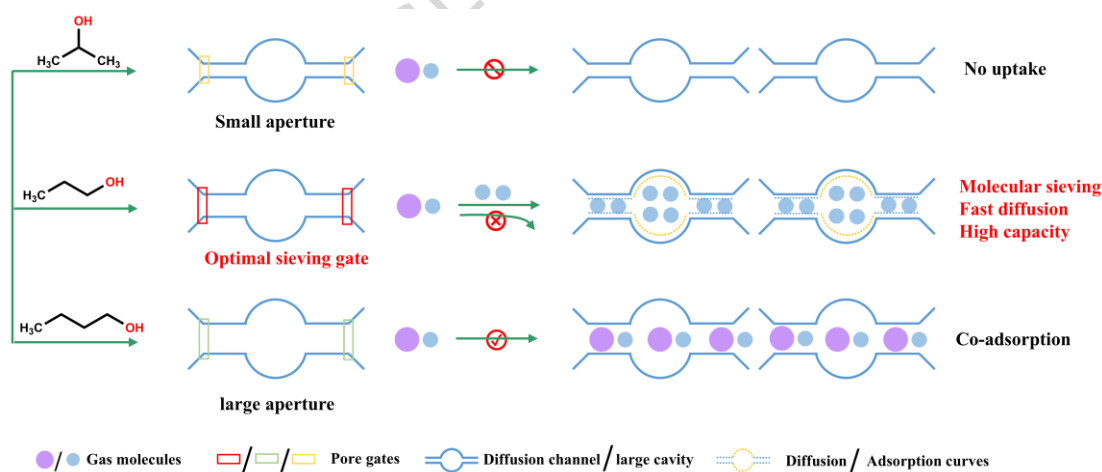


Fig. 1. Schematic illustration for optimal sieving gates. SDAs (Structure-Directing Agents) shape an optimal pore configuration in ZSTU-10, integrating precise sieving gates, interconnected diffusion channels and central microporous cavities.

Results

Synthesis and characterization

Colorless, rectangular prismatic crystals of ZSTU-10, ZSTU-11, and ZSTU-12 were synthesized via a solvothermal reaction involving zinc carbonate (ZnCO_3), 1,2,4-1*H*-triazole (TRZ), and oxalic acid in a solvent mixture of water (H_2O) and *n*-BuOH/*n*-PrOH/*i*-PrOH (Fig. 2a). Single-crystal X-ray diffraction (SCXRD) reveals that three isorecticular MOFs crystallize in the monoclinic $P2_1/c$ space group, with symmetrical building blocks comprising Zn nodes and organic linkers arranged systematically (Supplementary Table 1). In all three structures, each pentahedral Zn center is coordinated by three nitrogen atoms from three distinct triazolate linkers and two oxygen atoms from a single oxalate ligand. The planar conjugation of triazoles facilitates their preferential coordination with Zn, forming two-dimensional 2D layered $[\text{Zn}(\text{TRZ})]_n$. These layers are subsequently pillared by oxalate units to form a three-dimensional (3D) framework with a **dmc** topology (Supplementary Fig. 1), which is isostructural to the benchmark CALF-20 framework known for its high stability⁴³. The solvent-accessible voids for ZSTU-10, ZSTU-11 and ZSTU-12, as determined by the PLATON program, are 40.2%, 40.4% and 40.1%, respectively. The powder X-ray diffraction (PXRD) patterns of the as-synthesized samples match well with the simulated patterns, verifying their superior bulk phase purity and crystallinity (Supplementary Figs. 13-14). Notably, the PXRD patterns reveal a progressive shift of the peak for the (100) planes to lower 2θ values from ZSTU-10 to ZSTU-12. An in-depth examination confirmed that the template solvents induce subtle structural variations in bond angles and lengths. The quantitative $\angle \text{N-Zn-O}$ angles are 164.68° , 164.80° , and 165.09° , while the angles of adjacent triazoles are 98.60° , 99.37° and 97.48° for ZSTU-10, ZSTU-11, and ZSTU-12, respectively (Fig. 2b-d and Supplementary Fig. 5). These geometric variations effectively modulate the gate apertures and channel cross-sections.

SCXRD analysis, visualized via Connolly surface, demonstrates that ZSTU-10 features a center nanocage interconnected with surrounding diffusion channels. It possesses an optimal sieving gate of $4.4 \times 3.4 \text{ \AA}^2$ aligned with the crystallographic *c* axis (Fig. 2f), with an intermediate estimated cross-section area of $14.96 \text{ \AA}^2$, which is substantially larger than the minimum cross-section of C_3H_6 ($13.86 \text{ \AA}^2$) yet smaller than that of C_3H_8 ($15.58 \text{ \AA}^2$). By comparison, the gates in ZSTU-11 and ZSTU-12, with aperture dimensions of $4.9 \times 3.1 \text{ \AA}^2$, and $4.6 \times 2.9 \text{ \AA}^2$, either fails to recognize guests or blocks their entry (Supplementary Fig. 3-4). Since the gate aperture serves as the

dominant geometric determinant for sieving selectivity, these structural differences suggest that ZSTU-10 exhibits a potential size-selective sieving effect on C_3H_6 and C_3H_8 . Furthermore, the built-in large channels facilitate the rapid transport of guests into the central cavities, firmly locking them in place, which is crucial for prompting adsorbate diffusion and storage capacity. ZSTU-10 possesses a sufficiently large cavity with dimensions of $8.5 \times 6.3 \times 6.5 \text{ \AA}^3$, interconnected on all sides by diffusion channels $4.7 \times 3.0 \text{ \AA}^2$ (Fig. 2e and Supplementary Fig. 2). Such channel dimensions are sufficiently open to facilitate fast C_3H_6 diffusion, while the cage volume supports high packing density and thus high volumetric uptake. This comprehensive structural analysis suggests that ZSTU-10 possesses the optimal pore aperture and channel for C3 separation.

The structural robustness and phase identity of the ZSTU series were rigorously validated through complementary diffraction techniques and solvent-stability assays. To assess framework integrity upon activation, high-resolution neutron powder diffraction (NPD) was employed to characterize ZSTU-10/11/12. The NPD profiles of the as-synthesized and activated samples (ZSTU-10a/11a/12a) exhibit high structural uniformity with negligible deviation. (Supplementary Figs. 6-8 and Supplementary Figs. 11-12 and Supplementary Figs. 15-16). This structural invariance confirms that precise coordination environment and pore apertures are preserved without framework relaxation or collapse following the thorough evacuation of solvent molecules. Furthermore, systematic cross-solvent soaking experiments were conducted. Each material was immersed in the specific alcohols used for the synthesis of its isorecticular counterparts to evaluate potential phase transitions. The resulting PXRD patterns remained entirely unchanged (Supplementary Fig. 18), demonstrating that these frameworks do not undergo solvent-mediated interconversion. All ORTEP-style illustrations of ZSTU-10/11/12 were shown in the Supplementary Figs. 74-82. These results collectively confirm that the three isorecticular structures are independent stable states, whose specific pore metrics are inherent to the alcohol-templated assembly rather than reflecting meta-stable intermediates.

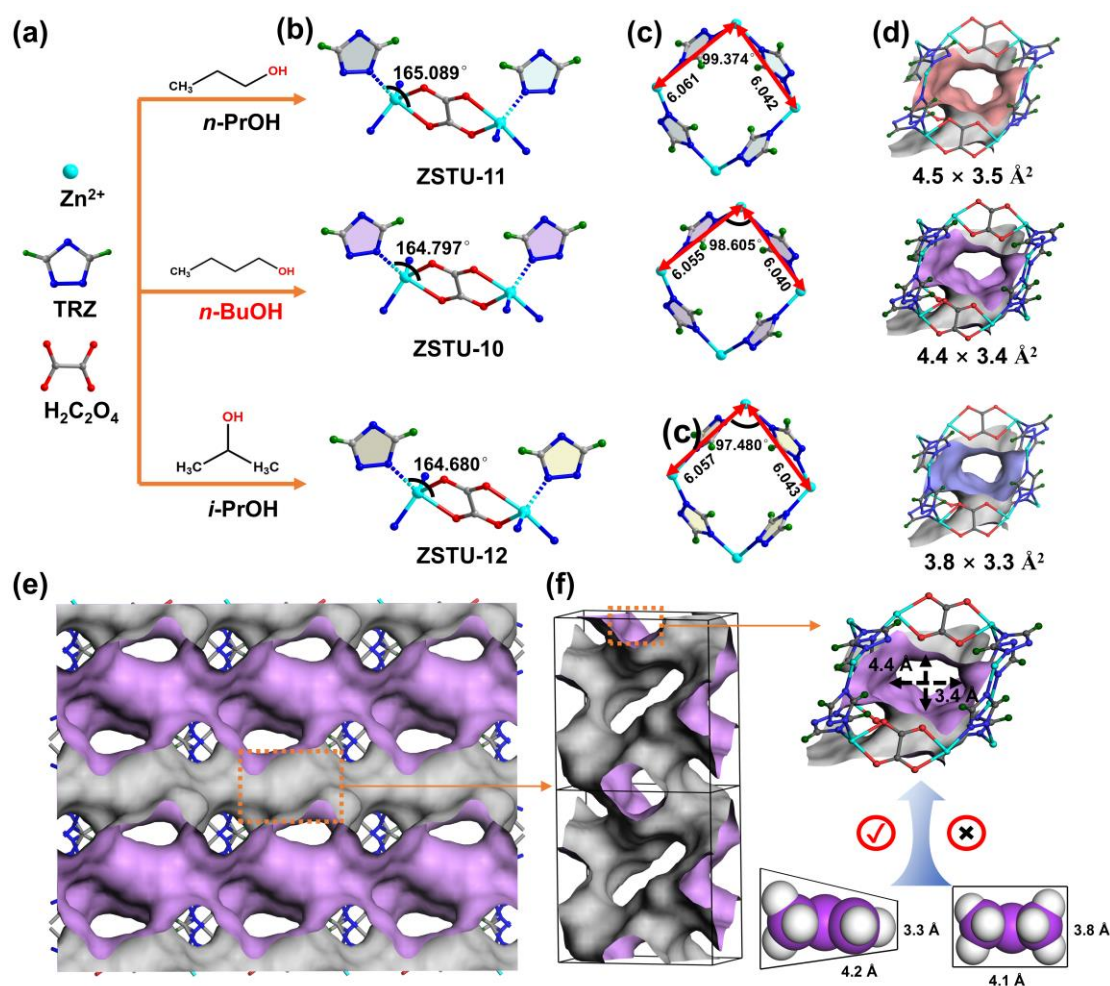


Fig. 2. Single-crystal structures of ZSTU series and pore architecture. (a) Synthesis procedure for the isostructural MOFs with solvent-templated n -butanol, n -propanol, and isopropanol. View of the topological structure (b) $\angle\text{N-Zn-O}$ angles along the c -axis and (c) $\angle\text{Zn-Zn-Zn}$ angles along the a -axis in ZSTU-10, ZSTU-11, and ZSTU-12 and (d) The corresponding pore windows parallel to the c -axis. (e) Cross-sectional view of the Connolly surface for the void in ZSTU-10 with an adopted probe radius of $0.8 \text{ \AA}$. (f) The large cavity surrounded by diffusion channels parallel to the b -axis (left) and the sieving gate and the molecular size of propylene (C_3H_6) and propane (C_3H_8) (the atoms of C_3H_6 and C_3H_8 are highlighted in purple) (right). Color code: Zn, light blue; C, gray; N, blue; O, red; H, green.

Porosity analyses and thermodynamic equilibrium adsorption

Thermogravimetric analysis indicates that all three MOFs maintain thermal stability up to $400 \text{ }^\circ\text{C}$. The initial weight loss steps correspond well with the evaporation temperatures of the

respective solvent templates (Supplementary Fig. 10). Subsequently, the permanent porosity of ZSTU-10, ZSTU-11, and ZSTU-12 was evaluated by carbon dioxide (CO₂) adsorption isotherms at 195 K and nitrogen (N₂) adsorption isotherms at 77 K (Supplementary Figs. 22-24). Both ZSTU-10 and ZSTU-11 exhibit type-I adsorption behaviors for CO₂ and N₂, whereas ZSTU-12 shows reversible type-I CO₂ adsorption isotherm but negligible N₂ uptake at 77 K, which is attributed to the kinetic restriction of N₂ diffusion through the narrower gate aperture in ZSTU-12. Based on N₂ adsorption isotherms at 77 K, the Brunauer–Emmett–Teller (BET) surface areas of ZSTU-10 and ZSTU-11 were determined to be 406.2 m² g⁻¹ and 419.8 m² g⁻¹, respectively (Supplementary Fig. 25). The corresponding total pore volume were determined to be 0.18 cm³ g⁻¹ and 0.249 cm³ g⁻¹. For ZSTU-12, its microporous nature is confirmed by a CO₂ uptake of 62.5 cm³ g⁻¹ at 195 K, with an apparent surface area of 180 m² g⁻¹ and a total pore volume of 0.097 cm³ g⁻¹. The pore size distribution for ZSTU-10, derived from the N₂ adsorption isotherm at 77 K, is narrowly centered at 3.9 Å based on Horváth-Kawazoe (H-K) model, consistent with an ultra-microporous gate-dominated framework.

Motivated by their distinct gate apertures and large cavities, we investigated the adsorption performance of C₃H₆ and C₃H₈ on three MOFs. Single-component adsorption isotherms of C₃H₆ and C₃H₈ on ZSTU-10, ZSTU-11, and ZSTU-12 were collected at 298 K and 1 bar (Figs. 3a-b and Supplementary Fig. 39). ZSTU-11 demonstrates substantial adsorption capacity for both C₃H₆ and C₃H₈, which is attributed to its larger pore aperture that lacks size discrimination. In contrast, ZSTU-12 exhibited minimal adsorption for either gas across various temperatures, indicating kinetic blockage due to its constricted pores. Notably, ZSTU-10 exhibited a steep uptake of C₃H₆ and quickly reached equilibrium at low pressure, while completely excluded C₃H₈. ZSTU-10 achieved a significant C₃H₆ volumetric uptake of 92.5 cm³ cm⁻³ (57.0 cm³ g⁻¹) at 0.1 bar and 298 K and reached a plateau at higher pressures. This exceptionally high low-pressure uptake is attributed to the synergistic effect of strong binding sites within the central cavity and rapid transport through diffusion channels, which was a common adsorption phenomenon observed in advanced adsorbents designed for high-affinity gas capture⁴⁴. At 298 K and 1 bar, The C₃H₆ adsorption capacity reaches 97.7 cm³ cm⁻³ (60.9 cm³ g⁻¹), reflecting its superior packing efficiency.

The excellent C_3H_6 uptake of ZSTU-10 surpasses many benchmark sieving materials, such as ZU-609 ($73.1/10.8 \text{ cm}^3 \text{ cm}^{-3}$)⁴⁵, Y-abtc ($63/35 \text{ cm}^3 \text{ cm}^{-3}$)³⁷, Co-gallate ($66.6/17.9 \text{ cm}^3 \text{ cm}^{-3}$)³⁸, JNU-3a ($87.5/18.7 \text{ cm}^3 \text{ cm}^{-3}$)³⁵, KAUST-7 ($61.9/10.1 \text{ cm}^3 \text{ cm}^{-3}$)⁴⁵⁻⁴⁸ at 1 bar and 0.1 bar (Fig. 3c and Supplementary Table 11). Due to negligible adsorption of C_3H_8 , ZSTU-10 possessed a significantly higher C_3H_6/C_3H_8 uptake ratio of 24.5, placing it among the top-performing materials (Supplementary Fig. 35)⁴⁹. Remarkably, based on the absorbed C_3H_6 amount and the experimental pore volume, the packing density of C_3H_6 within ZSTU-10 is estimated to be 633 g L^{-1} , which approaches the density of liquid C_3H_6 (608 g L^{-1}) at its boiling point of 225 K and exceeds gaseous C_3H_6 at 298 K and 1 bar (1.707 g L^{-1}) by a factor of approximately 370 times (Supplementary Fig. 34). These results indicate exceptionally efficient pore filling enabled by the optimized gate-channel-cavity architecture of ZSTU-10.

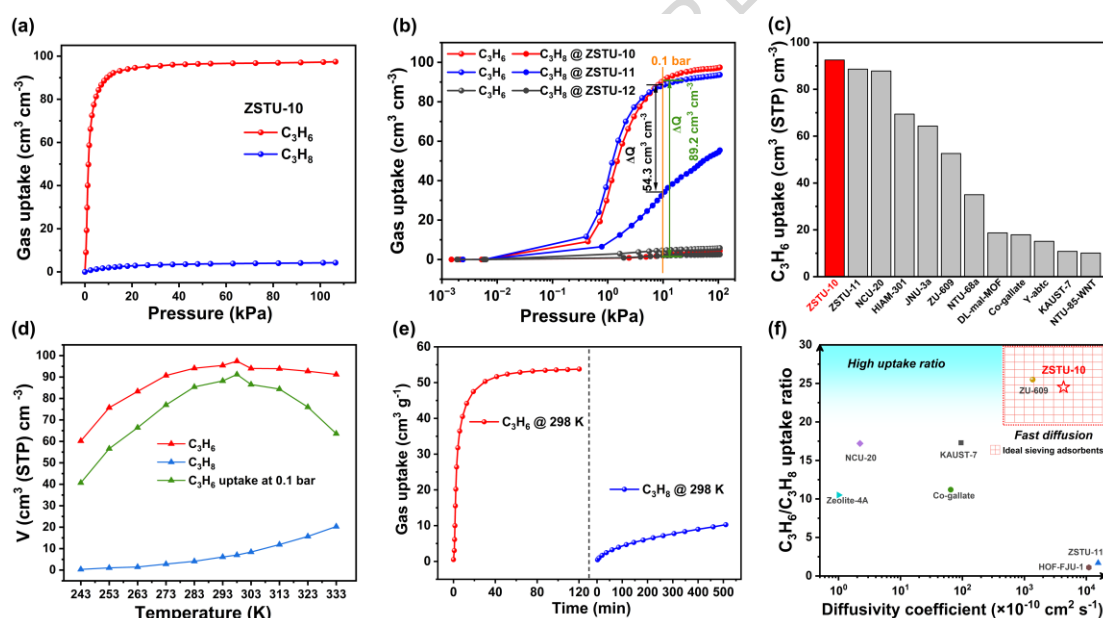


Fig. 3. Gas adsorption performances of C_3H_6 and C_3H_8 at 1 bar for ZSTU-10, ZSTU-11, and ZSTU-12. (a) Single-component adsorption isotherms of C_3H_6 (square, red) and C_3H_8 (circle, blue) at 100 kPa and 298 K for ZSTU-10. (b) The logarithmic plot of C_3H_6 and C_3H_8 on ZSTU-10 (red), ZSTU-11 (blue), and ZSTU-12 (black), differential values (Δq) of adsorption capacity for C_3H_6 and C_3H_8 at 0.1 bar. (c) Comparison of the C_3H_6 volumetric adsorption among other reported sieving MOFs at 0.1 bar and 298 K. (d) Temperature-dependent adsorption capacities (isobars) of C_3H_6 and C_3H_8 at 1 bar, and C_3H_6 uptake at 0.1 bar, measured across the temperature range of 243

K to 333 K. **(e)** Time-dependent sorption profiles of C₃H₆ (red) and C₃H₈ (blue) at 298 K and 1 bar. **(f)** Comparison of the diffusivity coefficient and C₃H₆/C₃H₈ uptake ratio among benchmark MOFs.

The ideal absorbed solution theory (IAST) was employed to evaluate the adsorption selectivity for an equimolar C₃H₆/C₃H₈ mixture, revealing an exceptionally high selectivity of 7×10^3 for ZSTU-10 (Supplementary Figs. 30-33 and Supplementary Table 11). Due to the ultralow capacity and the difficulty in achieving equilibrium for C₃H₈, fitting parameters of IAST selectivity for molecular sieving present certain uncertainties and errors that need to be acknowledged (Supplementary Tables 7-8). Encouraged by the outstanding low-pressure adsorption behaviors of ZSTU-10, Henry's constant of C₃H₆ and C₃H₈, along with Henry's selectivity were subsequently calculated based on the sorption isotherms across the entire pressure range with Langmuir fitting methods (Supplementary Fig. 36 and Supplementary Table 3)⁵⁰⁻⁵². The Henry's constant of C₃H₆ was estimated to be 174.3 mmol g⁻¹ kPa⁻¹ for ZSTU-10, and Henry's selectivity was determined to be 124, considering the low Henry's constant of C₃H₈ for ZSTU-10 (1.4 mmol g⁻¹ kPa⁻¹). This value far exceeds that of many C₃H₆-selective adsorbents, including DL-mal-MOF (5.86)⁴², and MAF-23-O (8.16)³⁹, and Mg-MOF-74 (18.7)⁵⁰ (Supplementary Fig. 37). Based on the adsorption isotherms at 313, 323 and 333 K, the isosteric heat of adsorption (Q_{st}) of C₃H₆ on ZSTU-10 was calculated to be 45.5 kJ mol⁻¹ at zero coverage using the virial method. For comparison, The Q_{st} of C₃H₆ on ZSTU-11 is 37.3 kJ mol⁻¹ according to C₃H₆ adsorption isotherms at 303, 313 and 323 K (Supplementary Figs. 26-29). The moderate adsorption enthalpy of ZSTU-10 indicates an energy-efficient regeneration process.

Adsorption kinetics

It is well established that the overall adsorption performance is governed not only by thermodynamic equilibrium factors, such as host-guest affinity and adsorption enthalpy, but also by the diffusion kinetics of guest molecules within the framework. An initial analysis of the structural features of the centered cage with diffusion channels in ZSTU-10 suggests that it is ideally configured to support the rapid kinetics of C₃H₆ molecules. To explore its kinetic advantage, time-dependent kinetic adsorption isotherms of ZSTU-10 and ZSTU-11 were measured from 283

K to 318 K at 1 bar (Supplementary Figs. 49-50). Remarkable differences in the diffusion coefficients of C_3H_6 and C_3H_8 were observed, with C_3H_6 diffusing much faster than C_3H_8 , leading to an impressive kinetic selectivity of 286 for ZSTU-10 (Supplementary Fig. 38 and Supplementary Figs. 51-52 and Supplementary Tables 4-5). To further quantify this distinction, the calculated diffusion coefficient of C_3H_6 in ZSTU-10 was $4.29 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$ at 298 K and 1 bar. This value is comparable to ZU-609 ($1.36 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$), and two orders of magnitude higher than other leading sieving materials such as Co-gallate ($6.47 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$), and KAUST-7 ($9.49 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$) (Fig. 3f and Supplementary Fig. 9 and Figs. 53-54 and Supplementary Table 6)⁵³. By combining this superior kinetics adsorption with the aforementioned thermodynamic equilibrium performances, ZSTU-10 achieves a more standard equilibrium-kinetic combined selectivity, which is up to 2×10^3 . Such a high selectivity significantly outperforms most benchmark MOFs, such as DL-mal-MOF (62.6)⁴², MAF-23-O (68.8)³⁹, Ni(AIP)(BPY)_{0.5} (48.3)⁵⁴ (Supplementary Table 10)⁵⁵.

In an unexpected finding, the static equilibrium adsorption displayed a pattern that deviated from conventional exothermic principles. As a result, pure-component adsorption/desorption isotherms were measured for C_3H_6 and C_3H_8 over a temperature range from 243 K to 333 K (Supplementary Figs. 40-48). With the increase of temperature, the results revealed a volcano-type trend for C_3H_6 in ZSTU-10. The C_3H_6 uptake initially rose from $60.2 \text{ cm}^3 \text{ cm}^{-3}$ to $97.3 \text{ cm}^3 \text{ cm}^{-3}$ and then decreased to $91.2 \text{ cm}^3 \text{ cm}^{-3}$, reaching a maximum at 298 K (Fig. 3d). In contrast, C_3H_8 adsorption does not reach equilibrium under the same conditions and shows an apparent increase with temperature, consistent with kinetically limited uptake. To probe the origin of this behavior, single-crystal X-ray diffraction data and in situ variable-temperature PXRD were collected at different temperatures (Supplementary Fig. 17 and Supplementary Table 1). Both analyses indicate that the framework of ZSTU-10 remains essentially rigid across the examined temperatures, and the key geometric parameters of the sieving gate ($4.4 \times 3.4 \text{ \AA}^2$) and diffusion channel show no discernible changes, indicating that these factors were not rationale for effect on adsorption (Supplementary Figs. 55-57). Similar temperature-dependent anomalies have been reported by Fernandez. et al⁵⁶ and Zhou. et al⁵⁷, leading to the inference that lower temperatures diminish the kinetic energy of guest molecules, hindering their ability to surpass the aperture energy barrier

despite favorable thermodynamics^{58,59}. Consistent with this interpretation, time-dependent adsorption isotherms demonstrated that the enhanced intrinsic kinetic energy of guest with rising temperature yield faster diffusion coefficients (Supplementary Table 6). It is suggested that the optimal configuration of the pore gate and diffusion channel collectively establish an optimal operating window around 298 K, where transport limitations are sufficiently alleviated while adsorption affinity remains high, thereby enabling simultaneous high thermodynamic and kinetic separation of C₃H₆/C₃H₈.

Host-guest structure studies

To elucidate the adsorption mechanism and host-guest interactions of C₃H₆ and C₃H₈ in ZSTU-10, we performed Grand Canonical Monte Carlo (GCMC) simulations at room temperature and 0.1 kPa, 10 kPa and 100 kPa, together with Dispersion-corrected density functional theory (DFT-D) calculations to evaluate the binding force of C₃H₆ molecules in ZSTU-10. At ambient temperature and 100 kPa, the density distribution maps indicate C₃ molecules are preferentially located near the oxalate ligands of ZSTU-10, proving that gas molecules diffuse into the cavity parallel to the c-axis of the pore window. Compared to the C₃H₈ density, the simulated C₃H₆ density was significantly higher at 0.1 kPa. Moreover, with the pressure increases, a dense packing of C₃H₆ molecules exhibits minimal variation between 10 kPa and 100 kPa, consistent with the experimentally observed steep low-pressure uptake (Supplementary Fig. 58). Furthermore, GCMC simulations revealed that two C₃H₆ molecules were adsorbed in the central large cavity, which is consistent with the static adsorption data. DFT-D calculations were conducted to quantify the adsorbent-adsorbate binding energy in ZSTU-10 (Supplementary Fig. 59). Corroborating the GCMC results, C₃H₆ molecules in ZSTU-10 primarily anchor to the oxygen atoms of the oxalate ligands, forming strong C-H $\cdots$ O interactions (2.701-2.987 Å). Additional weaker C-H $\cdots$ O interactions at around 3.933 Å are formed with distal oxygen atoms. Along the b-axis direction, C₃H₆ molecules interact with the triazole linkers via hydrogen-bonding interactions of d(C₃H₆ $\cdots$ TRZ) C-N $\cdots$ H at around 2.477-2.980 Å. The static binding energy of C₃H₆ at the binding sites was calculated to be 41.4 kJ mol⁻¹, which is in good agreement with the experimental Q_{st} value (45.5 kJ mol⁻¹). In comparison, the static binding energy of C₃H₈ was only 13.2 kJ mol⁻¹, revealing much

weaker binding forces. As depicted in Supplementary Fig. 62, C_3H_6 molecules located in the middle of the pore cavity in ZSTU-11, forming multiple $\text{C-H}\cdots\text{O}$ interactions (2.792-3.536 Å) and intermolecular $\text{d}(\text{C}_3\text{H}_6\cdots\text{TRZ})$ $\text{C-H}\cdots\text{N}$ (2.999-3.246 Å) interactions with the framework. The calculated static binding energy for C_3H_6 in ZSTU-11 is 42.4 kJ mol^{-1} , while the static binding energy for C_3H_8 is 20.3 kJ mol^{-1} , with the latter being higher than in ZSTU-10 (Supplementary Figs. 61-62). Moreover, electrostatic potentials (ESP) of the pore structure of ZSTU-10, ZSTU-11, and ZSTU-12 were also calculated by DFT-D calculations, and the differentiated framework energy was exhibited in Supplementary Figs. 63-65 (the corresponding energy order is $\text{ZSTU-11} > \text{ZSTU-10} > \text{ZSTU-12}$). The results reveal strong negative potentials near the oxygen atoms of oxalate ligands. Simulated results and calculated static binding energies are consistent with the experimental sorption phenomenon of ZSTU-10 and ZSTU-11.

To obtain direct structural insight into the adsorption behavior of ZSTU-10, high-resolution neutron powder diffraction (NPD) experiments were conducted to accurately establish the supramolecular binding forces between C_3D_6 molecules and the framework (Supplementary Fig. 11b). Two independent adsorption sites (site I and II) of C_3D_6 molecules were identified within the pore cavity constructed by four oxalate and eight triazole ligands, as shown in Figs. 4a-b and Supplementary Fig. 60). The occupancy of two C_3D_6 adsorption sites was refined to be 0.133 and 0.364 at sites I and II, respectively, indicating that each unit cell can accommodate two C_3D_6 molecules. This result corresponds to a volumetric uptake of $101.0 \text{ cm}^3 \text{ cm}^{-3}$ (STP), which is in excellent agreement with the experimental uptake from the C_3H_6 isotherm ($97.3 \text{ cm}^3 \text{ cm}^{-3}$) at 298 K and 1 bar. As depicted in Fig. 4, site II serves as the predominant sorption site and owns a higher occupancy, which was anchored by the surrounding oxalate ligands to form multiple $\text{C-D}\cdots\text{O}$ interactions (2.386-3.402 Å). Site I is located adjacent to the oxalate ligands and triazole ligands, binding to these ligands via $\text{C-D}\cdots\text{N}$ interactions (2.048-3.361 Å) and stronger $\text{C-D}\cdots\text{O}$ (2.222-3.501 Å) hydrogen bonds (Fig. 4c). In addition, a negligible framework deformation can be found when comparing the structure of activated ZSTU-10 with that of C_3D_6 -loaded ZSTU-10, confirming the structural rigidity of ZSTU-10. These results demonstrate that the precise pore window and expansive cavity are prerequisites for achieving effective molecular sieving and high-

density C_3H_6 packing, while the interconnected diffusion channels are essential for facilitating rapid guest transport.

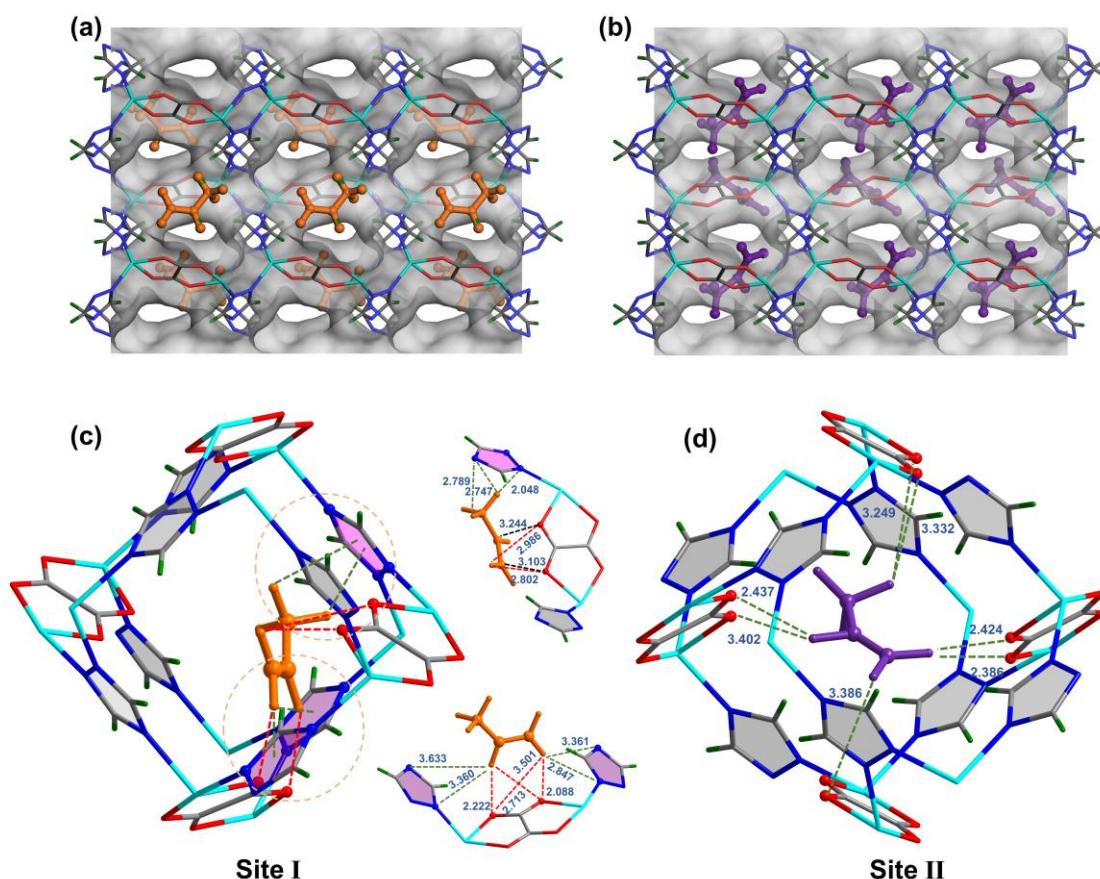


Fig. 4. Neutron diffraction crystal structure of $\text{ZSTU-10} \cdot 0.5\text{C}_3\text{D}_6$ and adsorption binding sites for C_3D_6 . Dense packing diagrams of the C_3D_6 -loaded crystal structure viewed along the c -axis of site I (a) and site II (b). The two preferential adsorption sites of C_3D_6 (site I, and site II) are represented in orange and purple, with two separate sites randomly distributed. (c) The occupancy of 0.133 for site I with close contacts with the framework structures, with $\text{C-D}\cdots\text{O}$ and $\text{C-D}\cdots\text{N}$ interactions highlighted in red and green, respectively. (d) The occupancy of 0.364 for site II with mainly preferential $\text{C-D}\cdots\text{O}$ interactions.

Breakthrough experiments

Motivated by the superior separation capacity of ZSTU-10 in single-component adsorption/desorption performance, we further conducted binary $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ ($v/v=50/50$) column breakthrough experiments, in which equimolar binary mixtures flowed over the sample bed of MOFs with different flow rates at 298 K (Supplementary Fig. 66). Dynamic breakthrough

experiments revealed that both ZSTU-10 and ZSTU-11 are capable of separating C_3H_6 from an equimolar $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixtures (Fig. 5a and Supplementary Fig. 71). However, the separation efficiency distinguishes the two materials. Notably, at a total flow rate of 3 mL min^{-1} , the breakthrough curves for the equimolar $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixture show that C_3H_8 elutes first, and the co-adsorption interval is short (4.2 min g^{-1}). After the overlap ends, a clear time window is observed in which C_3H_6 is retained while C_3H_8 has already eluted, enabling a wide separation window with mass-normalized retention time of approximately 16.6 min g^{-1} for high-purity C_3H_6 collection (Fig. 5a). The dynamic C_3H_6 capacity of ZSTU-10 was calculated to be 2.76 mmol g^{-1} based on $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ breakthrough adsorption curves, surpassing that of ZU-609 (1.64 mmol g^{-1})⁶⁰, KAUST-7 (0.65 mmol g^{-1})³⁷, Y-abtc (1.26 mmol g^{-1})³⁷ and Co-gallate (0.98 mmol g^{-1})³⁸ (Fig. 5b and Supplementary Fig. 67). At a higher flow rate of 5 mL min^{-1} , ZSTU-10 maintains a limited co-adsorption interval (8.6 min g^{-1}), whereas ZSTU-11 exhibits a substantially longer co-adsorption interval (15 min g^{-1}) under comparable conditions (Supplementary Figs. 68 and 72). Besides, the retention time of $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ for ZSTU-11 below the co-adsorption interval reveals the poor separation efficiency. The shorter overlap for ZSTU-10 indicates reduced co-elution and thus improved dynamic separation performance for equimolar $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixtures. The excellent breakthrough performance of ZSTU-10 prompted us to explore C_3H_6 purity and productivity, the desorption experiments were conducted through helium gas sweeping (10 mL min^{-1}) at 448 K (Supplementary Fig. 69). Correspondingly, the purity and productivity of the preferentially adsorbed C_3H_6 was 99.1% and 37.5 L kg^{-1} (Figs. 5e-f and Supplementary Fig. 70). These metrics exceed those of reported benchmark MOFs with sensational performance, such as for KAUST-7 (16.3 L kg^{-1} , 90% purity)⁴⁶ and Y-abtc (1.3 L kg^{-1} , 90% purity)³⁷ (Supplementary Table 9). Five consecutive breakthrough adsorption/desorption measurements for an equimolar $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixture were carried out at the flow rate of 5 mL min^{-1} . The results revealed consistent retention times, no obvious loss of capacity, and a sustained C_3H_6 dynamic capture at around 3.7 mmol g^{-1} (Fig. 5c and Supplementary Fig. 68), confirming the excellent regenerability and cycling stability of ZSTU-10. Overall, these dynamic results demonstrate the practical potential of ZSTU-10 for $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ separation. Because C_3H_8 adsorption is strongly suppressed by molecular sieving,

regeneration primarily involves desorbing C_3H_6 from the high-affinity sites, which can reduce unnecessary energy expenditure associated with co-adsorbed impurities.

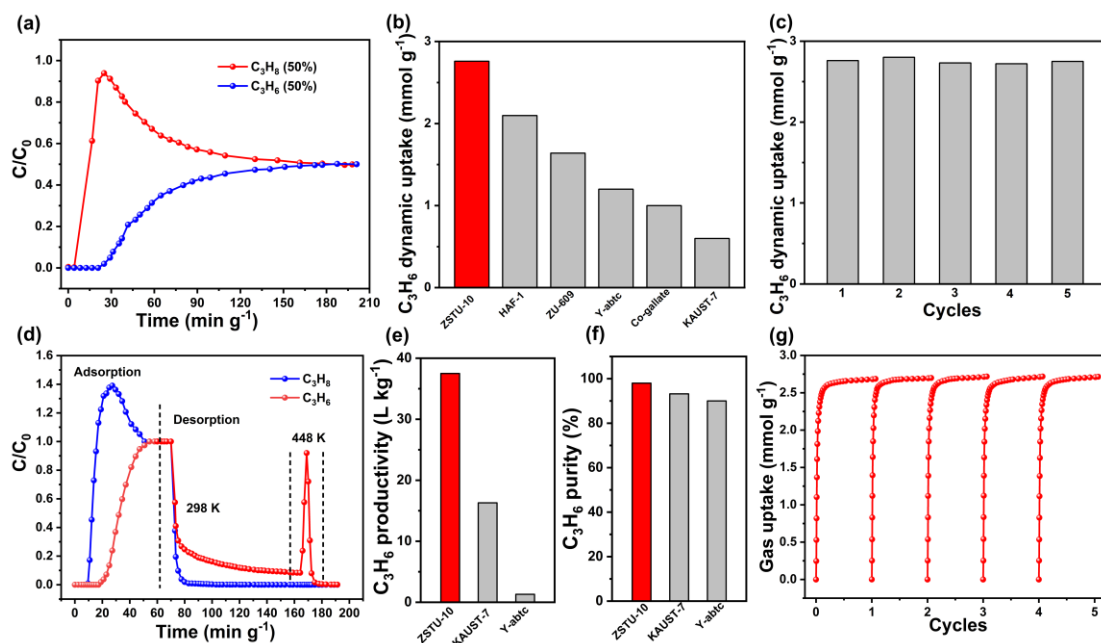


Fig. 5. Dynamic breakthrough experiments of ZSTU-10. For calculation and comparative analysis, all breakthrough time parameters are reported as mass-normalized breakthrough time (min g⁻¹). (a) Dynamic column curves in ZSTU-10 for C_3H_6/C_3H_8 (50/50) with a flow rate of 3 mL min⁻¹. (b) Comparison of the dynamic C_3H_6 uptake of ZSTU-10 with other reported C_3H_6/C_3H_8 sieving materials. (c) Dynamic uptake of cyclic tests on ZSTU-10 for C_3H_6/C_3H_8 mixtures. (d) Dynamic adsorption breakthrough curves at a flow rate of 5 mL min⁻¹, followed by desorption curves under Argon gas with a sweeping flow rate of 10.0 mL min⁻¹. The desorption profile shows two distinct stages: an initial purge at 298 K, followed by thermal desorption at 448 K (resulting in the concentrated C_3H_6 peak). (e) Comparison of C_3H_6 productivity with well-known MOFs. (f) Comparison of C_3H_6 purity among other famous sieving MOFs. (g) Cycling adsorption-desorption curves on ZSTU-10 at 298 K and 1 bar.

Given the demanding conditions prevalent in practical industrial production, we investigated the stability of ZSTU-10 under a range of relevant environments. The synthesized sample was exposed to air for 15 d and immersed in water for a week. The PXRD patterns indicate that ZSTU-10 can maintain substantial structural integrity without phase change or loss of crystallinity. Stable adsorption capacities of C_3H_6 for ZSTU-10 can be preserved after treatment with water at different

time intervals (Supplementary Fig. 19). Furthermore, this material was soaked in different chemical environments for a week, such as organic solvents (DMF, MeOH, EtOH, CH₂Cl₂, ACN and acetone) and acid and alkaline solutions (pH = 2, 4, 10, 12). In all cases, the samples exhibited PXRD patterns identical to the simulated one, demonstrating exceptional chemical stability (Supplementary Figs. 20-21). Five consecutive single-component adsorption-desorption isotherms were conducted at 298 K and 1 bar, which consistently showed high levels of C₃H₆ adsorption capacity on ZSTU-10 (Fig. 5g and Supplementary Fig. 72). Additionally, the synthesis of ZSTU-10 uses inexpensive, readily available raw materials and demonstrates potential for scalability. Combined the advantages of good stability, low production cost, and the ability to simultaneously exclude C₃H₈ while efficiently adsorbing C₃H₆ renders ZSTU-10 a promising candidate for industrial application.

Discussion

Employing a solvent-template strategy, we synthesized three isomorphic ultra-microporous MOFs (ZSTU-10/11/12) with systematically tuned pore apertures. Among them, ZSTU-10 integrates combines localized sieving gate with expansive diffusion channels connected to a central storage cavity, enabling complete C₃H₈ exclusion while facilitating selective C₃H₆ recognition with fast diffusion kinetics. This gate-channel-cavity architecture achieves an optimized thermodynamic-kinetic separation of C₃H₆/C₃H₈ mixtures, yielding exceptional selectivity up to 2×10^3 under ambient conditions. ZSTU-10 simultaneously breaks off inherent challenges in balancing static thermodynamics and kinetics, with ultra-high C₃H₆ adsorption capacity, fast kinetic, and regenerable host-guest interactions. Dynamic breakthrough experiments confirmed a high productivity of 37.5 L kg⁻¹ per adsorption-desorption cycle. DFT calculation and NPD studies further elucidate preferential binding of C₃H₆ via multiple weak interactions with framework O/N sites, consistent with the experimentally measured adsorption thermodynamics. The robust framework stability, recyclability, and tolerance to moisture and acidic/basic environments of ZSTU-10 underscore the potential of diffusion-confinement MOFs for challenging gas separation applications. Our work provides compelling evidence for the utilization of highly efficient molecular sieve separation adsorbents, featured by high adsorbate's adsorption capacity and rapid

diffusion kinetic, thereby paving a new engineering platform for demanding gas separation.

Methods

Synthesis of ZSTU-10

ZnCO₃ (0.627 g, 5 mmol), H₂C₂O₄ (0.45 g, 5 mmol), and 1,2,4-triazole (TRZ, 1.727 g, 25 mmol) were added to the mixture of H₂O (15 mL) and N-butyl alcohol (*n*-BuOH) (15 mL) in a 100 mL Teflon-lined reaction vessel. The mixed solution was stirred for an hour and reacted in an oven at 453 K for three days. After cooling to room temperature naturally, the resulting colorless crystals were collected by filtration and washed thoroughly with methanol. Finally, the product was dried in air overnight. To assess the potential for scale-up, we optimized the synthesis conditions. We confirmed that ZSTU-10 can be synthesized with high purity and crystallinity using a reduced molar ratio of 1:1:5, significantly lowering ligand consumption. Based on this optimized ratio and laboratory reagent prices, the estimated raw material cost is approximately \$50.1 kg⁻¹ (Supplementary Fig. 73), which serves as a promising baseline for future industrial economic evaluation.

Synthesis of ZSTU-11 and ZSTU-12

ZSTU-11 and ZSTU-12 were synthesized following the same procedure as ZSTU-10, except that *n*-BuOH was replaced by N-propyl alcohol (*n*-PrOH) or Isopropyl alcohol (*i*-PrOH), respectively.

Single-component sorption experiments

Prior to gas adsorption tests, approximately 200 mg of samples in the sample cell were activated under a dynamic vacuum at 423 K overnight to remove the guest molecules until the pressure stabilized. After each single-adsorption measurement, the sample cell was degassed at 333 K under vacuum for 0.5 h. C₃H₆ and C₃H₈ adsorption isotherms were collected from 243 K to 333 K on an automatic volumetric adsorption instrument (Micromeritics 3-Flex analyzer, 3-Flex Version 5.02). For each pressure point, an equilibrium time of 6 min was applied. Data were recorded only after a stability criterion was met: the system pressure changed by less than 0.1% over a consecutive 2 min period. The experimental adsorption-desorption temperatures were controlled by the ice bath (273 K) and a constant water bath (298 K). The Brunauer-Emmett-Teller (BET) specific surface areas were analyzed using nitrogen adsorption isotherms at 77 K. The 195 K CO₂ was performed

in the mixtures of dry ice and acetone. The low temperatures from 243 K to 263 K were controlled by the mixtures of dry ice, ethanol, and ethylene glycol. The data visualization uses OriginPro 2024.

Dynamic breakthrough experiments

Fixed bed breakthrough experiments were carried out in a packed stainless steel column (10 mm $\times$ 12 cm) at 298 K and 1 bar. The breakthrough column was filled with activated samples (3.6 g) at ambient conditions. A stream of Argon was introduced into the column to purge the adsorbents for 15 h at 423 K in a temperature controller of the thermostat before the breakthrough experiments. The total inlet flow rate was controlled by calibrated mass flow controllers (MFCs) and was kept constant throughout each breakthrough experiment. The breakthrough column experiments were conducted using the gas mixture of C₃H₆/C₃H₈ (v/v, 50/50) with a stable flow rate of 5 mL min⁻¹ at 298 K, and the outgas of the column was monitored by a gas chromatograph (GC-2014C Shimadzu) equipped with a thermal conductivity detector for continuous gas analysis. When both C₃H₆ and C₃H₈ reached equilibrium, the desorption experiments were conducted through high-purity Ar (99.999%) with a flow rate of 10 mL min⁻¹ at 298 K, and then heated up to 448 K to sweep. Five consecutive breakthrough adsorption–desorption cycles were performed to evaluate the cycling stability of ZSTU-10.

Data availability

The authors declare that the data supporting the findings of this study are available within the article and Supplementary Information. Crystallographic data for the structures in reported this article have been deposited at the Cambridge Crystallographic Data Centre, under deposition numbers CCDC 2424397 (ZSTU-10), 2424398 (ZSTU-11), 2424396 (ZSTU-12), 2424393 (ZSTU-10a), 2424394 (ZSTU-11a), 2424395 (ZSTU-12a), 2424392 (ZSTU-10·C₃D₆), 2424400 (ZSTU-10b), and 2424399 (ZSTU-10c).. Copies of the data can be obtained free of charge via <https://www.ccdc.cam.ac.uk/structures/>. The PXRD patterns, TGA curves, sorption tests, gas adsorption data, Rietveld refinement of powder X-ray diffraction tests and breakthrough tests that support the findings of this study are provided as a Source Data File (ref. 61. Xue, W (2026): Source data of ZSTU-10, ZSTU-11, and ZSTU-12 that support the findings of this study.xlsx.

Source Data file has been deposited in Figshare under accession code (<https://doi.org/10.6084/m9.figshare.30018499>). A reporting summary for this article is available as a Supplementary Information file.

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Author contributions

X.W. performed the experiments associated with material synthesis, data analysis and prepared the first version of the manuscript. L.B. refined the single-crystal X-ray diffraction data and revised the manuscript. L.B., S.T., T.K., and P.M. performed the neutron powder diffraction experiments and related data analysis. T.Z. performed the gas adsorption experiments. H.C. and L.L. measured the fixed-bed breakthrough tests. X.W., T.Z. and J.L. analyzed the structural data. P.M., R.-B.L., B.C. and J.G. directed and supervised the project. J.G. conceived and designed the research. All authors discussed the results and commented the manuscript.

Competing interests

The authors declare no competing interests.

Editorial Summary:

Researchers report a metal-organic framework with “gate-channel-cavity” for the energy-efficient separation of propylene. By combining sieving gates with a large diffusion channel, the material achieves record-fast diffusion and high capacity for industrial use.

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